MCKV Institute of Engineering

Project Title: **Employee Performance Analysis**

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A Project Presentation for Industrial Training at

Ardent Computech Pvt. Ltd

Project Introduction

This project focuses on developing an Employee Performance Dashboard using Power BI to analyze workforce data, evaluate employee performance, and understand distribution across departments. It also helps identify key factors affecting attrition and supports data-driven decision-making.

Data Visualization

HR datasets transformed into actionable workforce insights

Interactive Dashboard

Dynamic visuals with slicers and cross-filtering

Strategic Decision-Making

Data-driven HR decisions supported by visual analytics

Project Objectives

The objective of this project is to examine key employee metrics to gain insights into performance and organizational trends.

It aims to identify patterns that can improve workforce efficiency and reduce employee attrition.

01

Workforce Overview

To understand how employees are distributed across departments and locations, along with gender distribution and count of total Employee.

02

Performance Analysis

To assess employee productivity based on key performance metrics like Projects handled, working hours etc.

03

Attrition Insights

To identify patterns and factors contributing to employee turn-over, salary correlations, overtime effects, and tenure patterns

04

Strategic Planning

Data-driven decisions for retention, training programs, and workforce optimization.

Key Dashboard Features

Interactive Visualizations

Bar charts, donut charts, line charts, maps, and matrix tables for comprehensive data exploration

Dynamic Slicers

Age group, department, job role, work-life balance, and gender filters for targeted analysis

Theme Toggle

Dark mode and light mode switch for improved visual comfort and accessibility

Multi-Page Layout

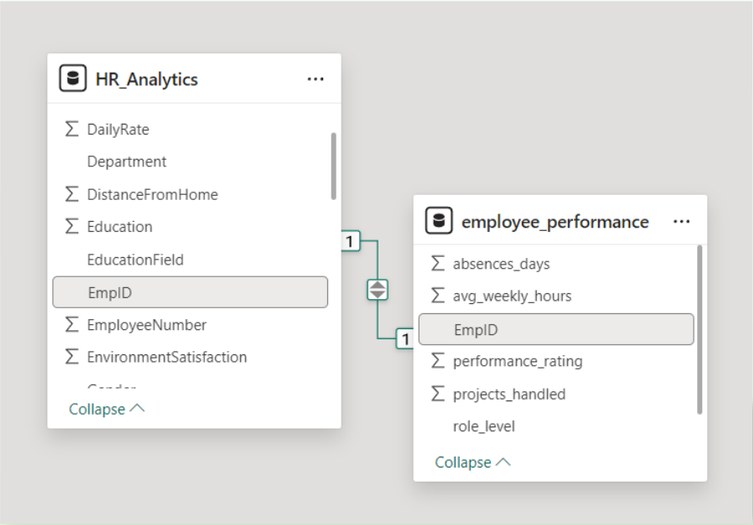
Employee overview, performance metrics, and attrition analysis across dedicated report pages

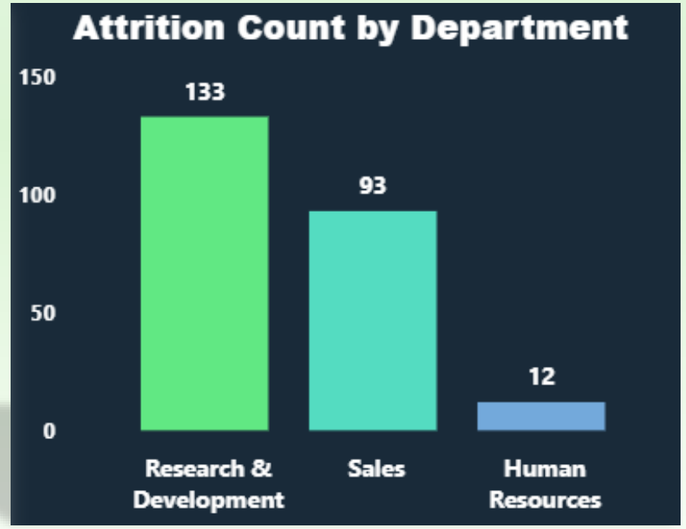
KPI Indicators

Card visuals displaying total employees, average income, years at company, and attrition rate

Cross-Filtering

Interactive filtering between visuals to analyze relationships across different variables





Implementation Methodology

Data Collection

HR Analytics and Employee Performance datasets from Kaggle with 1,480 employee records

Data Cleaning

Power Query Editor used to remove duplicates, verify data types, and ensure consistency

Data Modeling

Two tables connected via EmpID with one-to-one relationship for accurate analysis

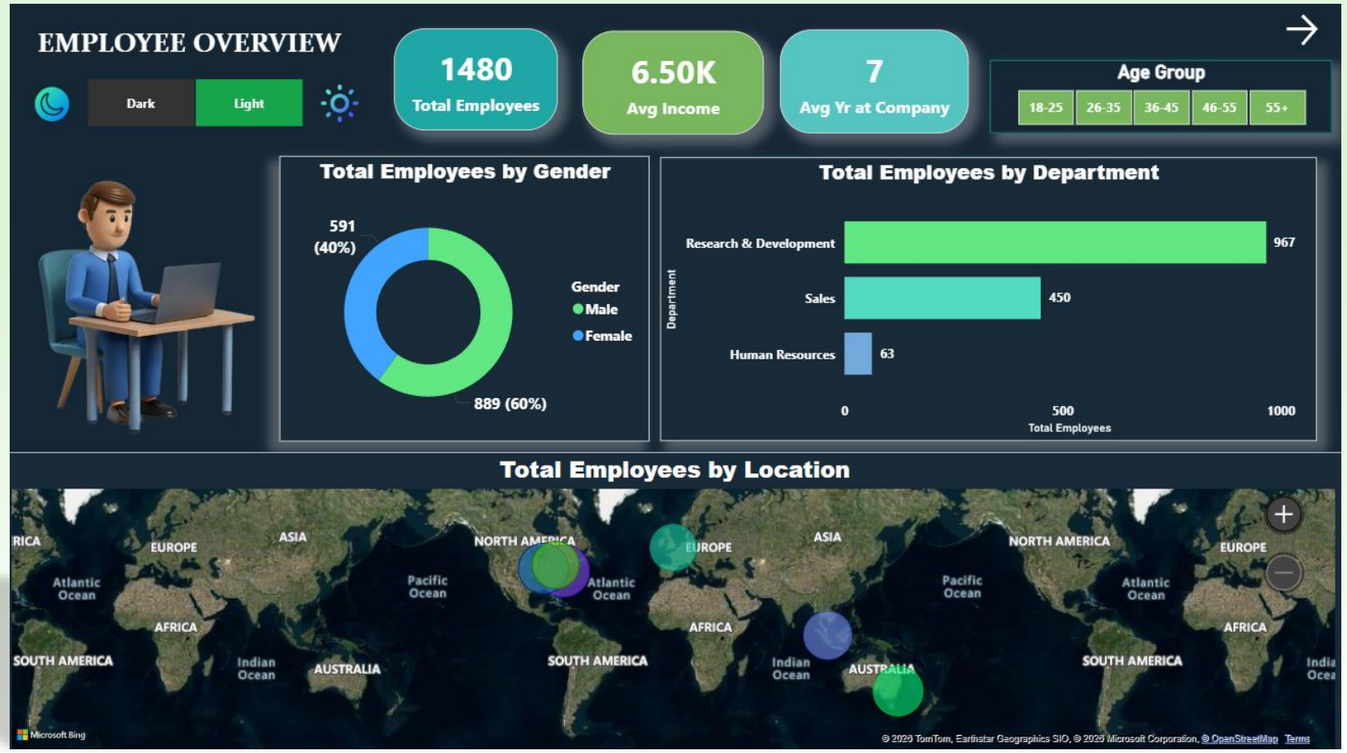
Visualization

DAX measures and Visualization for KPIs, charts, and interactive dashboard elements

Key Insights from Analysis

Workforce Profile

- 1,480 total employees with 60% male and 40% female distribution.
- Research & Development department has highest employee count.
- Average tenure of 7 years indicates workforce stability.
- Average monthly income of $6,500 across organization.
- The Map chart reveals a geographically diverse workforce.



Performance Trends

- 6,668 total projects handled reflecting productivity levels.
- Average 50 weekly working hours indicating workload.
- Senior employees contribute more significantly to projects.
- Training shows positive relationship with performance ratings.
- Absenteeism alone is not a reliable indicator of performance.

EmpID	Total Projects	Performance_rating	Avg_weekly_hours	Absences_days	MonthlyRate
RM001	1	1	55	17	25233
RM002	2	4	36	13	9724
RM003	2	2	61	20	8059
RM004	4	1	36	17	13493
RM005	5	5	53	5	13556
RM006	2	5	38	1	19305
RM007	6	4	59	18	18420
RM008	5	5	37	15	8018
RM009	7	2	38	10	26820
Total	6668	4358	73779	14453	21161722

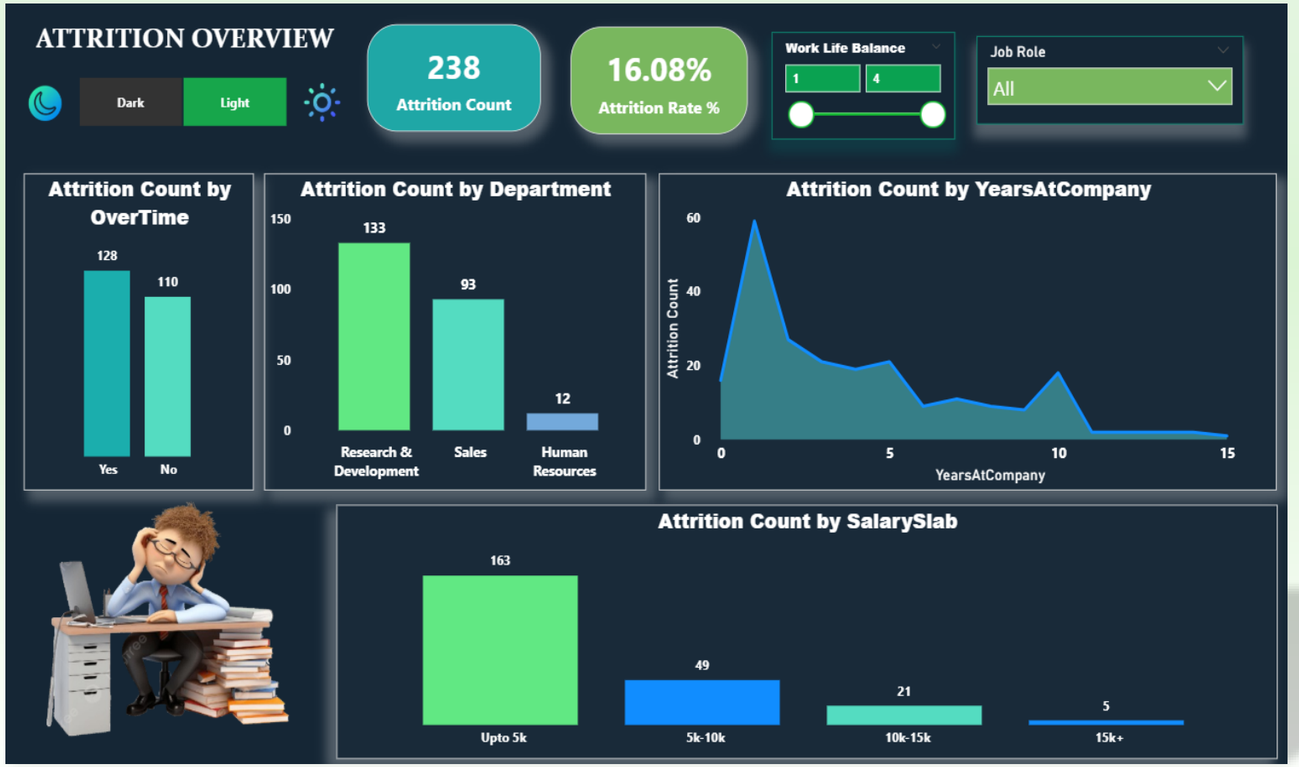
Attrition Analysis Findings

238
Employees Left
Total number who departed organization

16.08%
Attrition Rate
Overall turnover percentage

Early Years
High Risk Period
Most attrition occurs during initial employment years

Overtime
Key Factor
Employees working overtime more likely to leave



Others

- Lower salary groups show higher attrition compared to higher salary groups.
- Research & Development department experiences highest turnover.

Advantages and Limitations

Advantages

- ❑ Interactive and user-friendly dashboard for easy data analysis.
- ❑ Dynamic filtering using slicers for better data exploration.
- ❑ Comprehensive view of employee performance, distribution, and attrition in a single dashboard.
- ❑ Efficient use of Power BI features like DAX, visuals and data modeling.
- ❑ Flexible design that can be adapted to other domains also.

Limitations

The dataset used in this project is static (Kaggle) and does not support real-time updates. The project does not include predictive analytics or advanced machine learning techniques, and the desktop-based dashboard limits accessibility and sharing while requiring a Power BI setup for full functionality.

Conclusion and Future Scope

Conclusion

The Employee Performance Dashboard successfully demonstrates how data analytics transforms raw employee data into actionable workforce insights. Through interactive visualizations, organizations can monitor performance, identify attrition patterns, and support data-driven HR decisions.

Future Enhancements

Real-Time Integration
Connect to live HR databases for monitoring metrics in real time

Predictive Analytics
Machine learning models to predict employee attrition risk

Cloud Deployment
Power BI Service for secure access and collaboration across departments

Mobile Optimization
Responsive design for monitoring insights on mobile devices

Thank You

Special thanks to MCKV Institute of Engineering & Ardent Computech Pvt. Ltd., for guidance and support throughout this training project. Thank you so much for listening.

~ By Sayantani Dalui